

Salah Chikhi

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PUBLICATIONS & MANUSCRIPTS

Adaptive Scheduling Pipeline for Multi-Instance Asynchronous Reinforcement Learning.

Salah Chikhi*, Luis Henrique Zuin Ruiz*, Entong Li, Li Zeng. *NeurIPS Main Track 2026 (accepted)*. [OpenReview](#)

Last-Iterate Performance of Gradient Descent and Relaxed Proximal Point via s -Composability.

Salah Chikhi. *Submitted to INFORMS Journal on Optimization, 2026*. [arXiv:2609.13765](#)

Taylor Representations for Scalable Model-Free Reinforcement Learning in Networked Markov Decision Processes.

Salah Chikhi, Abdelhaq Chaoui, Asuman Ozdaglar, Saurabh Amin. *Submitted to ICLR 2027*.

An Analytical Framework for the Levine Hats Problem: New Strategies, Bounds and Generalizations.

Clément Bouquet*, Salah Chikhi*, Timothé Charles, Yanghao Zhou, Eric Wang. *Preprint, 2025*. [arXiv:2508.01737](#)

ε -Sufficiency for Experiment Design under Ellipsoidal Uncertainty.

Abdelhaq Chaoui, Salah Chikhi, Asuman Ozdaglar, Saurabh Amin. *Manuscript targetting ICML 2027*.

A Causal Model to Explain Complexity of Topics and Its Empirical Link with Corpus Velocity.

Salah Chikhi, Keyvan Attarian, Baptiste Arnaudo, Charles-Albert Lehalle. *Manuscript, 2026*.

EDUCATION

University of Oxford | *MSc in Mathematics*

Oxford, UK | 2026–2027

- MSc student at Oxford's Master in Mathematics which emphasizes research and preparation for doctoral studies.
- Relevant Coursework: Theories of Deep Learning, Continuous Optimization, Random Matrix Theory, Advanced Topics in Statistical Machine Learning, Perturbation Theory, Stochastic Analysis and PDEs, Further Functional Analysis, Algorithmic Foundations of Learning

École polytechnique | *Ingénieur Polytechnicien (MSc)*

Palaiseau, France | 2023–2027

- GPA: 3.94/4.0 — Master's in Applied Mathematics at France's most selective STEM institution.
- Relevant Coursework: Optimization for AI, Deep Learning, Optimization & Control, Statistics and learning, Advanced Probability, Stochastic Calculus, Advanced Statistical Learning, Complex analysis and differential calculus...

Lycée du Parc | *Classes Préparatoires MPSI-MP* (BSc equivalent)*

Lyon, France | 2021–2023

- GPA: 4.0/4.0 — Intensive two-year program in mathematics, physics, and computer science; selected for the *MP** (Advanced Maths & Physics) track for France's most selective competitive entrance exams..

RESEARCH EXPERIENCE

Massachusetts Institute of Technology | *Visiting Student, EECS*

Cambridge, MA | March 2026 – Present

- **Project 1:** Developed a scalable reinforcement learning framework for high-dimensional networked MDPs with local interactions, deriving localized Taylor expansion based value function approximations with theoretical guarantees for policy evaluation and policy optimization.
- The proposed method outperforms the state-of-the-art Random Fourier Features baseline on tasks like LQR while requiring weaker assumptions on the transition kernel and significantly fewer learnable parameters.
- **Project 2:** Studying data informativeness in the context of optimization for unobserved functions.
- Derived an efficiently computable spectral certificate $\rho(D)$, proved that $\rho(D) \leq \varepsilon^2$ guarantees an at most ε -regret solution to LP programs, characterized the observation subspace which minimizes $\rho(D)$ through an eigenspace construction, and showed polynomial decay of $\rho(D)$ with the size of the dataset D on a broad class of problems.

Huawei Technologies | *AI Research Intern*

Shenzhen, Ox Horn Campus | June–September 2025

- Designed and implemented **ASTP**, a novel RL post-training pipeline for LLMs with 200B+ parameters, eliminating GPU/NPU idle times, scheduling skewness, and padding bubbles inherent to classical hybrid pipelines.
- ASTP combines a mathematically-grounded task scheduling policy (optimized via Metropolis–Hastings and dynamic programming) with a state-of-the-art asynchronous rollout–training dependency, achieving a **3.3 $\times$** end-to-end efficiency gain and **2 $\times$** rollout throughput improvement; **now deployed in production reducing telecom LLM training from >3 months to 2 months**.
- Concurrently contributed to a second team, developing novel methods to preserve LLM reasoning capabilities during supervised fine-tuning on low-reasoning datasets.
- Received a return offer; associated research **accepted at NeurIPS'26 main track**.

The Levine Hats Conjecture | *Combinatorics, Measure Theory, Probability*

2024

- Team of five (École polytechnique), supervised by Dr. Lucas Gerin. Developed a measure-theoretic framework recasting strategies as Lebesgue-measurable functions on $[0, 1]^{n-1}$, yielding a unified integral expression of the optimal winning probability V_n for both finite and infinite stacks.
- Constructed K_5 , a new recursive strategy achieving the conjectured optimum $7/20$; proved the first rigorous geometric lower bound $V_{2,h} \geq 7/20 - C \cdot 256^{-h/5}$, improving all prior bounds and resolving conjectured inequalities on $(V_{2,h})_{h \geq 1}$.
- Introduced and entirely solved the continuous Levine game (uncountable stacks): proved the optimal winning probability is exactly $1/2$, and derived optimal responses to arbitrary fixed strategies.
- **Preprint:** [arXiv:2508.01737](https://arxiv.org/abs/2508.01737)

Causal Model for Topic Complexity and Corpus Velocity | *Structural Causal Models, Information Theory, NLP* 2025

- Team of three, supervised by Dr. Charles-Albert Lehalle. Developing a principled complexity metric $\mathcal{C}(M_i)$ for conceptual markers in textual corpora, grounded in Structural Causal Models: a marker is complex if its articulation presupposes a rich constellation of co-occurring concepts.
- $\mathcal{C}$ is defined as the average pairwise lift (i.e. $\exp(\text{PMI})$) within semantically coherent clusters recovered by a hierarchical pipeline (UMAP + DBSCAN for clustering, PC algorithm for intra-cluster causal graph recovery), with theoretical guarantees on both steps.
- Established a robust stylized fact on millions of financial and economic documents from 155 sources: publication velocity is a **decreasing function of complexity** across 21 thematic clusters and across individual publishers—with a closed-form optimal source-selection rule as a downstream application for RAG corpus governance.
- Manuscript in preparation.

COMPETITIONS & AWARDS

International Mathematics Competition for University Students (IMC) – 2024: Bronze Medal**International Physics Olympiad (IPhO) – 2021:** Honorable Mention in the French Selection Test**École polytechnique Mathematics Olympiad Selection Test:** Ranked 2nd/80**National Coach – Algerian Team, International Olympiads in AI (2025–present):** Team achieved **2 Gold & 2 Silver medals**, ranking **1st and 2nd** in AOAI. Silver medal in IOAI.

TEACHING

École polytechnique | *Teaching Assistant*

Palaiseau, France | September 2024 – January 2025

- Led tutoring and review sessions for 2nd-year Bachelor students in topology of normed vector spaces and differential/integral calculus in several variables.

Lycée Saint-Exupéry | *Mathematics Teacher*

Mantes-la-Jolie, France | September 2023 – April 2024

- Taught 100+ underprivileged high school students across STEM disciplines; prepared 3 undergraduate classes for France's competitive grandes écoles entrance exams.
- Coached 6 gifted students for the *Concours Général de Mathématiques*, France's most prestigious high school mathematics competition.
- Served on the organizing committee and jury of a regional mathematics competition spanning 3 institutions, from problem-set design through final grading.

SKILLS & INTERESTS

Languages: French (native), English (C2 — TOEFL 116/120)**Programming:** Python, Java, PyTorch, scikit-learn, Ray, vLLM, pandas, $\text{\LaTeX}$ **Interests:** Basketball, Volleyball, E-Sports (former semi-professional; peak rank **17th/40M** worldwide in Overwatch), Travelling & Culture